COMP0051 Coursework 2024/2025
CN: NWVL2

1. Introduction

This paper explores the performance of leveraged algorithmic trading strategies on SPTL Exchange-Traded Fund (ETF) time-series data, representing long-term U.S. Treasuries performance. We implement mean-reversion and forecasting strategies and optimise model selection through a train-validation-test split. The Effective Federal Funds Rate (EFFR) is used as a proxy for the risk-free rate, reflecting the Federal Reserve's monetary policy stance.

2. Data Preprocessing

Time series data was sourced from Yahoo Finance for the SPTL-ETF Closing Price and the Federal Reserve Bank of St. Louis (FRED) for the EFFR rate. The period of interest starts from the 1st of January 2023 to the 31st of December 2023, including 251 observations. The latter full-sample undertakes a 50% train, 30% validation and 20% testing set to effectively evaluate performance on unseen out-of-sample data, avoiding any data leakage.

The following figures showcase full-sample time series data of SPTL returns and EFFR rates.

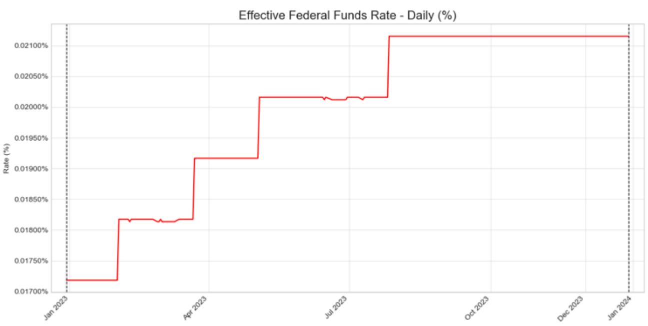

Figure 1: The daily EFFR rate from January 2023 to December 2023. The series displays incremental step sizes, reflecting the Federal Reserve's rate hikes during the period of interest.

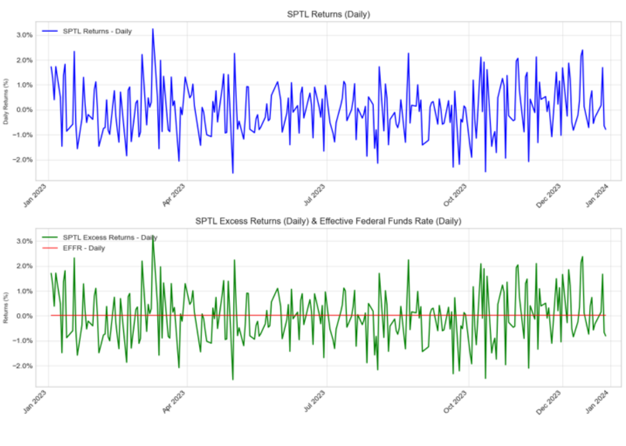

Figure 2: Full Sample SPTL Daily Returns and Excess Return per unit of SPTL with daily EFFR overlay.

We define excess returns per unit of SPTL:

$$r_t^e = r_t - r_t^f ,$$

Where $r_t = {}^{\Delta p_t}/_{p_t}$, denoting daily SPTL Returns and r_t^f as the daily EFFR Rate.

Figure 2 showcases the minimal difference between the SPTL return and excess return series, which can be explained by the small fluctuations in the daily EFFR rate (overlayed in red). Figure 1 confirms increases in the daily EFFR rate whilst showcasing the relatively small magnitude of these increments.

3. Methodology

Algorithmic trading strategies are defined as sets of rules for buying and selling assets, which in turn generate a signal. Their primary goal is to optimise portfolio returns according to different risk tolerances. Dynamic strategy returns at time t (r_t^s) are defined by the signal generated at the previous timestep t-1 (s_{t-1}) and the excess return per unit of SPTL at time t (r_t^e).

$$r_t^s = s_{t-1} \cdot r_t^e$$

In our context, $\{\theta_t\}_{t=1}^{T}$ is the time series data for the strategy's dollar value positions in the SPTL-ETF. We thus limit position sizing at time t through the following constraint:

$$|\theta_t| \leq V_t \cdot L$$

The initial condition is represented by our initial capital $V_0 = \$100{,}000$ and the leverage limit $L = 10$. Any unused portion is assumed to be held in a money market account earning the daily risk-free rate, which we calculate through the Annualised EFFR rate.

3.1. Mean Reversion Strategy

Mean Reversion is a phenomenon describing how an asset's return time series reverts back to its long-term average with short fluctuations in between. Vasicek models in 1977 the instantaneous short rate - here proxied by the daily EFFR- as a mean-reverting process [1]. Since excess returns are defined by subtracting this proxied rate from SPTL returns, it follows that any departure from their equilibrium should tend to revert.

This mean reversion effect is visible in *Figure 2,* where SPTL returns and excess returns (blue and green, respectively) tend to oscillate around zero mean, further reinforcing our choice.

We define our mean-reverting strategy as follows, in binary format:

$$s_t = \begin{cases} 1, & \text{if } r_t < MA_t^{(\tau)} \\ -1, & \text{if } r_t > MA_t^{(\tau)} \\ 0, & \text{otherwise} \end{cases}$$

With $MA_t^{(\tau)} = \sum_{i=0}^{\tau} w_k r_{t-k}$ a simple moving average of past SPTL percentage returns over a look-back window of size τ. In this configuration; $s_t = 1$ corresponds to a long position; the current return is below the look-back expected return, which suggests an incoming upward adjustment. The other case; $s_t = -1$, corresponds to a short position and indicates a potential future downward adjustment.

In our implementation of this strategy, we let deviation magnitudes drive our position and clip the signal between -1 and 1 inclusive, such that:

$$s_t = \text{clip}(MA_t^{(\tau)} - r_t, -1, 1)$$

We implement a simple grid search approach to find the optimal look-back window size. For each window size, we evaluate strategy performance on both the training (50%) and validation (30%) samples.

window	train Sharpe	val Sharpe
10	4.705963	0.270911
12	4.534200	0.245257
8	5.001028	0.216802
15	4.011440	-0.014976
20	3.813040	-0.480260

Table 1: *Train and Validation Sharpe Ratios for different look-back periods (i.e. fixed window sizes)*

The lookback window size yielding the highest Sharpe Ratio on the validation set (0.2709) is selected (i.e. best window size = 10 observations)

3.2. ARIMAX Forecasting Strategy

We use an Auto-Regressive Moving Average with Exogeneous Variables (ARIMAX) model for our second strategy. This is a forecasting strategy, our signal is built upon comparing the ARIMAX prediction at time t+1 with the current value at time t.

A crucial aspect for ARIMAX implementation is the stationarity of the time series data it fits. An Augmented Dickey-Fuller (ADF) test applied to the full sample SPTL closing price time series p_t fails to reject the null-hypothesis of non-stationarity, with an ADF statistic of -1.254 and a p value of 0.6499.

By contrast, an ADF test on the SPTL daily returns time series $\{r_t\}$ showed strong stationarity with ADF statistic = -17.98 and p-value <<< 0.001. This initial analysis determined r_t as the response variable. An ARIMAX model of order (1, 0, 0) on SPTL returns $\{r_t\}$ takes the form:

$$r_t = c + \emptyset_1 \cdot r_{t-1} + \beta \cdot x_t + \epsilon_t$$

With x_t the exogenous variables and ϵ_t is a Gaussian white noise error term. The one-step daily return forecasts are computed with:

$$\hat{r}_{t+1} = \hat{c} + \hat{\emptyset}_1 \cdot r_t + \hat{\beta} \cdot x_t$$

The ARIMAX forecasting strategy can now be explicitly defined as follows:

$$s_t = \begin{cases} 1, & \text{if } \hat{p}_{t+1} > p_t \\ -1, & \text{if } \hat{p}_{t+1} < p_t \\ 0, & \text{otherwise} \end{cases}$$

with $\hat{p}_{t+1} = p_t \cdot (1 + \hat{r}_{t+1})$ as the next day's close STPL price forecast.

The following model selection pipeline is implemented:

1. We first fit pure ARIMA (p, d, q) models (i.e. without any exogenous regressors) on the training sample, choosing the order that minimises the AIC. With an AIC of -786.4 the selected ARIMAX order was (1, 0, 0). Not accounting for exogenous variables in this stage avoids overfitting due to the limited amount of full sample data we have and the large pool of candidate regressors (i.e. exog variables) we want to select for optimisation.

2. Stationarity analysis of candidate exogenous variables is undertaken for adequate ARIMAX implementation.

Variable	Description	Series ID / Ticker
Y5	5-Year U.S. Treasury yield	DGS5
Y10	10-Year U.S. Treasury yield	DGS10
Y30	30-Year U.S. Treasury yield	DGS30
2Y10Y	10-Year minus 2-Year yield spread	T10Y2Y
5Y30Y	30-Year minus 5-Year yield spread	DGS30 – DGS5
t_premium	Kim–Wright 10-Year term premium	THREEFYTP10
inflation	10-Year Breakeven Inflation Rate	T10YIE
move_daily	ICE BofA Merrill Lynch MOVE Index	^MOVE

Table 2: Exogenous Variables considered for model selection.

Except for the *{move_daily}* series, for which we take the percentage change, all remaining series are first differenced. ADF tests on these transformed exogenous time series confirm stationarity, with all p-values <<< 0.001.

3. With the ARIMAX order fixed from the training sample, we implement a grid search of the optimal set of exogenous variables. For each combination of up to three exogenous variables, we generate one-step return forecasts to evaluate Sharpe performance on the validation set.

features	train Sharpe	val Sharpe	train MaxDD	val MaxDD
y30, 2Y10Y, t_premium	1.932855	2.598483	-0.849281	-0.566564
2Y10Y, 5Y30Y	0.604336	2.349622	-1.172826	-0.721454
y5, t_premium	0.899181	2.054131	-2.255335	-0.996115
y10, y30, 2Y10Y	1.987224	2.042448	-2.255335	-0.577775
y10, y30, t_premium	1.809641	1.930353	-1.848712	-1.972413

Table 3: *Train and Validation set Sharpe Ratios for different Exogenous Regressor Combinations (with additional metrics – i.e. Maximum Drawdown)*

The features leading to the highest validation Sharpe are selected: the 10-Year U.S. Treasury yield time series, the spread time series between the 10-Year U.S. Treasury yield and the 2-Year U.S. Treasury yield and the term premium time series on 10-Year Zero Coupon Bonds. All three time series are **first differenced** as highlighted point 2 of the "model selection pipeline". Additionally, for the same features, we observe the smallest Maximum Drawdown, indicating that the selected exogenous variables could provide good returns and controlled downside risk.

4. Finally, we fit the final ARIMAX model over the full in-sample data (training and validation sets) to check residuals are coherent with Gaussian white noise.

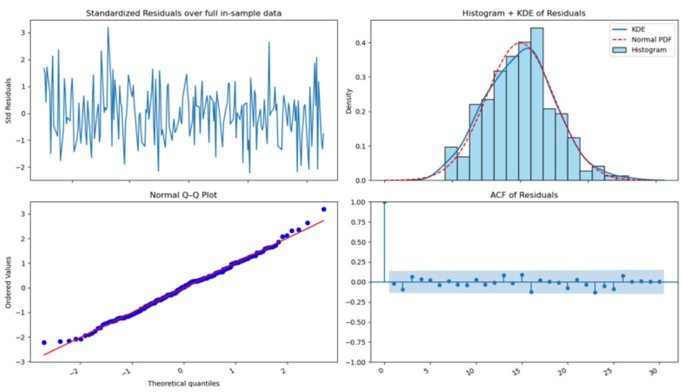

Figure 3: *Residual Diagnostics for the final ARIMAX(1,0,0) model - Top-left: standardised residuals; top-right: histogram (blue) with overlaid Normal PDF (red dashed); bottom-left: Normal Q-Q plot; bottom-right: ACF of residuals.*

The analysis in Figure 3 shows that residuals behave like white noise and pass Ljung-Box & ARCH-LM checks. We can now proceed to genuine out-of-sample testing on completely unseen data using the held-out test set.

3.3. Performance Metrics

The following metrics are used to evaluate the performance of our dynamic trading strategies.

3.3.1. Sharpe Ratio

The Sharpe Ratio allows to measure the risk-adjusted performance of a given strategy. We use the annualised Sharpe defined as:

$$Sharpe = \frac{E[r_t] \cdot 252}{\sigma[r_t] \cdot \sqrt{252}}$$

With $\{r_t\}$ as the total return of the strategy.

3.3.2. Maximum Drawdown

This metric quantifies the maximum observed loss from the peak to the trough of our strategy. It is a percentage of the peak value:

$$Maximum\ DrawDown = \max\left\{\frac{V_{peak} - V_{trough}}{Vpeak}\right\}$$

3.3.3. Calmar Ratio

Combines the Sharpe Ratio and Maximum Drawdown to account for strategy performance per unit risk

$$Calmar\ Ratio = \frac{Annualised\ Return}{Maximum\ Drawdown}$$

4. Strategy Results

For each strategy, we refit the entire strategy on the combined 50% train and 30% validation sets, which make up the full in-sample of our data (i.e. to plot full in-sample performance). We then run the strategy on the completely unseen test set (i.e. to plot the full out-of-sample performance).

4.1. Profit and Loss

The total daily Profit and Loss (PnL or ΔV_t^{total}) - is determined by the formula:

$$\Delta V_t^{total} = \Delta V_t + \Delta V_t^{cap},$$

Where ΔV_t represents the daily change in value of our position in the "risky" asset and ΔV_t^{cap} the daily change in the money-market account growing at the risk-free rate r_t^f.

The daily P&L of the "risky" asset is given by:

$$\Delta V_t = \left(\frac{\Delta p_t}{p_t} - r_t^f\right) \cdot \theta_t$$

Whereas the daily PnL of the money-market account, with M_t, the margin requirement:

$$\Delta V_t^{cap} = (V_t^{total} - M_t) \cdot r_t^f$$

5. Mean Reversion Results

After fitting the mean reversion model on the full in-sample data, with the selected optimal look-back window size (=10) we observe the following PnL performance.

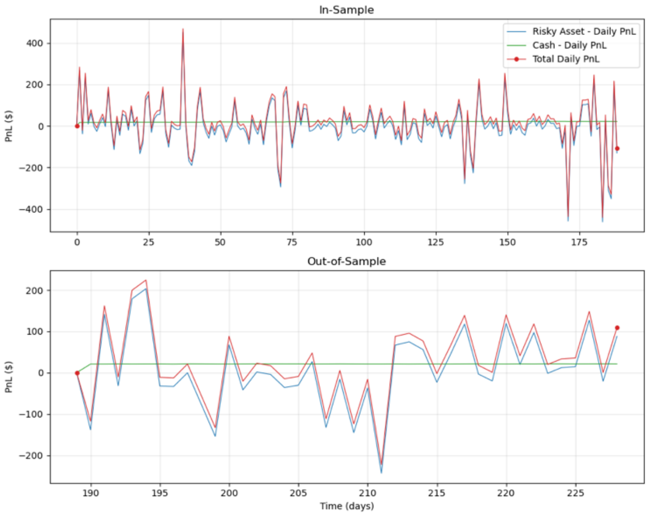

__Figure 4:__ Mean Reversion Strategy daily PnL for the in-sample data (top) and out-of-sample data (bottom)..

The presence of a non-binary signal in the mean reversion strategy allows for unused capital to be invested in the money market account. The in-sample panel of Figure 4, shows how the total daily PnL and the daily PnL in the risky asset seem to oscillate around the mean, showcasing short-term reversion of in-sample profits over the window length.

The out-of-sample period has somewhat increased volatility but the strategy continues to generate positive net PnLs on most days.

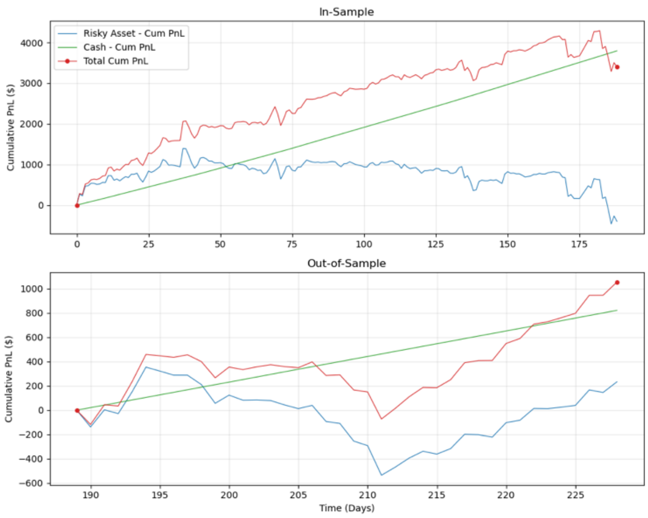

__Figure 5:__ Mean Reversion Strategy cumulative PnLs for the in-sample data (top) and out-of-sample data (bottom)

In the out-of-sample period, the total cumulative PnL dips slightly before recovering to about $1000 at the end of the year.

	Sharpe Ratio	Max Drawdown	Calmar Ratio	Total Return
In-Sample	2.757	-0.234	0.192	3395.328
Out-of-Sample	4.647	-1.160	0.058	1053.531

__Table 4:__ Final Mean-Reversion Strategy In-Sample Metrics (80%) and Out-of-Sample (20%) Metrics- i.e. total returns are in $.

6. ARIMAX Results

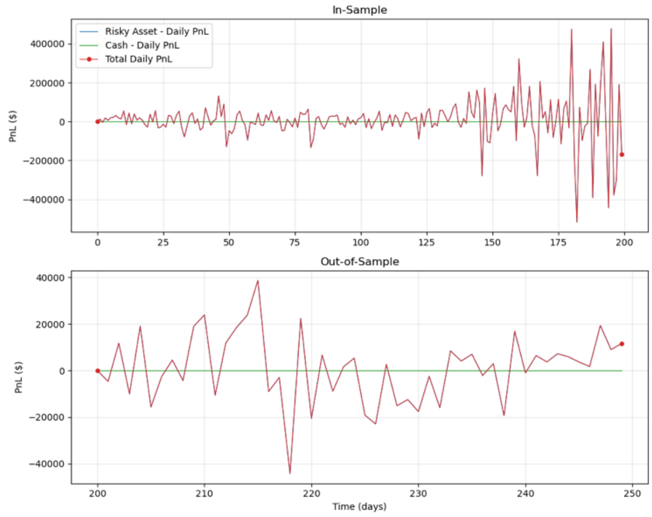

__Figure 6:__ ARIMAX Strategy daily PnLs for the in-sample data (top) and out-of-sample data (bottom).

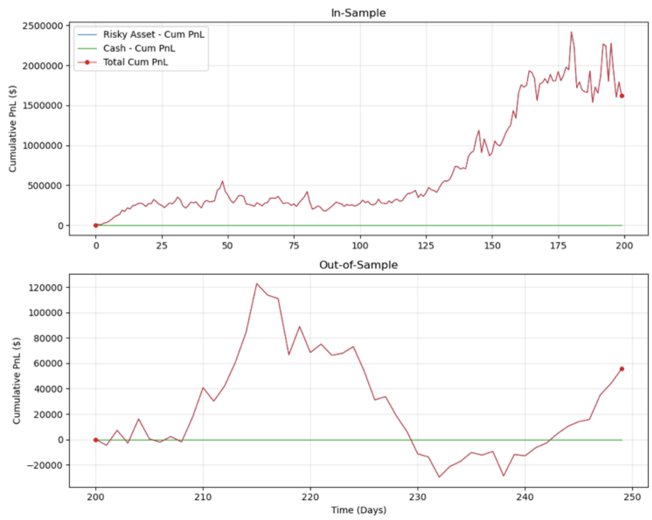

__Figure 7:__ ARIMAX Strategy cumulative PnLs for the in-sample data (top) and out-of-sample data (bottom).

	Sharpe Ratio	Max Drawdown	Calmar Ratio	Total Return
In-Sample	3.067	-0.679	7.237	1621676.176
Out-of-Sample	2.170	-1.417	2.775	55645.228

__Table 5:__ Final ARIMAX model In-Sample Metrics (80%) and Out-of-Sample (20%) Metrics - i.e. total returns are in $.

7. Conclusion

The ARIMAX Strategy outperformed mean-reversion in terms of Total Gross Return. The higher Max Drawdown observed on out-of-sample ARIMAX compared to out-of-sample mean-reversion is explained by the non-binary nature of the mean reversion signal which allows for smaller gains but better downside control.

However, the numerous exogenous features evaluated in the ARIMAX implementation have proved to be the differentiating factor between the two models. For a lower sharpe ratio

It remains that by optimising model hyperparameters and feature selection on an additional validation set has proven to ensure a robust implementation of the described strategies. The results on purely out-of-sample data seem promising enough to keep developing the ideas outlaid in this report. It is worth noting that excessive feature selection was avoided to reduce risks of overfitting the in-sample data (due to the limited length of data available – i.e. one year's worth).

Bibliography

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